Technology Stack (Architecture & Stack)

Date	26 June 2026
Team ID	
Project Name	OptiCrop : Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example:

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap 5, JavaScript	Provides a responsive, user-friendly interface for entering agricultural parameters and displaying crop recommendations across devices.
2	Backend / Server-Side	Python 3.10+, Flask	Flask offers a lightweight and efficient framework for integrating the machine learning model, processing user requests, and generating predictions quickly.
3	Database / Data Storage	CSV Dataset, Pickle (.pkl)	The agricultural dataset is stored in CSV format for model training, while the trained machine learning model is saved as a Pickle (.pkl) file for fast prediction without retraining.
4	Cloud / Hosting / Deployment	Flask Development Server (Local Deployment)	Enables easy local testing, debugging, and demonstration of the application before future deployment to cloud platforms such as AWS or Azure.

5	Version Control & CI/CD	Git.Github	Facilitates source code management, version control, collaboration, and backup of project files throughout development.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook	Scikit-learn is used for training and evaluating ML models; Pandas and NumPy handle data preprocessing; Matplotlib and Seaborn support data visualization; Jupyter Notebook is used for experimentation and model development.